

**Boston 311 Service Requests: A Neighborhood Level Analysis of Government
Responsiveness and Equity (2015 – 2019)**

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Introduction:

This report continues an ongoing analysis of the City of Boston's 311 non-emergency service request system. The goal of the report is to determine whether Boston resolves constituent service requests with equal speed and diligence regardless of the racial composition and socioeconomic status of the neighborhood in which the request originates. Four research questions guide the whole project - RQ1 asks whether 311 response times and overdue rates differ systematically across Boston neighborhoods based on income, poverty rate, racial/ethnic composition, and whether these differences persist after controlling for department, submission channel, year, and season. RQ2 asks whether this neighborhood-based gap widened or narrowed over the 2015 – 2019 period. RQ3 asks whether, when controlling for specific complaint types and case urgency, the positive association between neighborhood poverty and 311 response delays persist and if this resolves the counter-intuitive income effect. RQ4 asks whether the equity gaps in response time and overdue rate manifest uniformly across all municipal departments or are driven by specific operational units. After controlling for department, submission channel, year, and season, both neighborhood poverty rate and racial/ethnic minority share were significant predictors of longer response times and higher odds of an overdue case. On the other hand, neighborhood median household income showed a small positive association with delay. A chi-square test of independence is added as a distribution-appropriate method to check the descriptive overdue-rate gap without relying on regression assumptions. These analyses move the project from establishing a neighborhood-based gap exists towards explain where the gap comes from. It also explains how large the gap really is once complaint mix and department are accounted for, and whether it is concentrated in few operational units or spread evenly across the city of Boston.

Analysis:

The analytic dataset is the merged and cleaned 1,031,658 case files produced from the five annual 311 extracts (2015 – 2019) and the BPDA/Census neighborhood profile. The first method was multiple linear regression, and it was used to answer RQ1: response time (hours) was regressed on department group, submission channel, year, season, neighborhood median household income, poverty rate, and percent minority. The reference categories were department = Inspectional Services, channel = Mobile App, year = 2015, and season = Fall. Method 2 utilized binary logistic regression to answer RQ1 as well and this is the primary binary-outcome test of this research question. Case overdue status (1 = overdue) was regressed on the same predictor set, with coefficients reported as odds ratios with 95% Wald confidence intervals. Method 3 used extended multiple linear regression to answer RQ3, and it is a base linear model regressed response time on poverty rate and neighborhood income tier alone. A controlled model added case compliance stats (on-time and overdue) and the complaint type (the case's "type" field with over a hundred categories). Comparing poverty-rate and income-tier coefficients across the base and controlled specifications tests whether the neighborhood effect on response time is essentially asking if where you live still affects how fast complaints get answered even after accounting for what type of complaint it is. An interaction linear regression method was utilized to help answer RQ4 which regressed response time on poverty rate, income tier, department, and the full set of poverty by department and income tier by department interaction terms. The interaction coefficients test whether the size and direction of the income and poverty effect on response time depends on which municipal department is handling the case. Lastly, a Pearson chi-square test of independence was conducted on the 2 by 2 contingency table of neighborhood income tier (higher vs. lower income) by case status (on-time vs overdue).

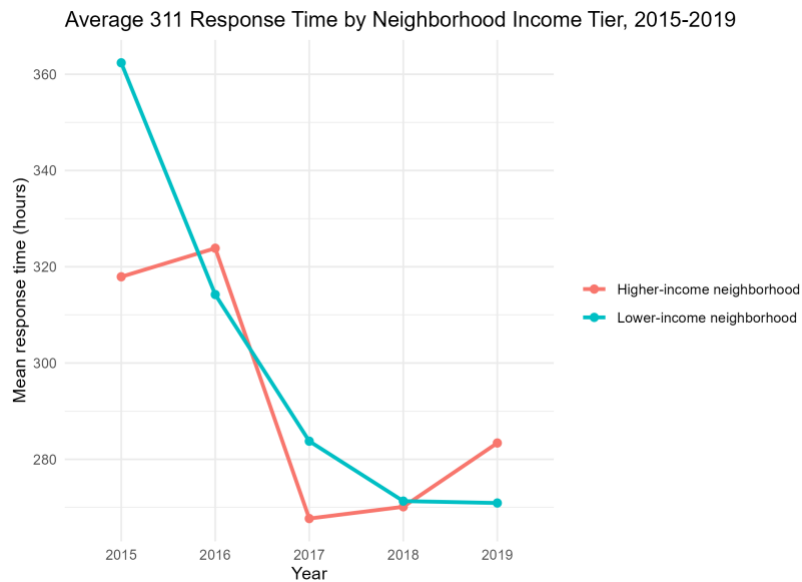
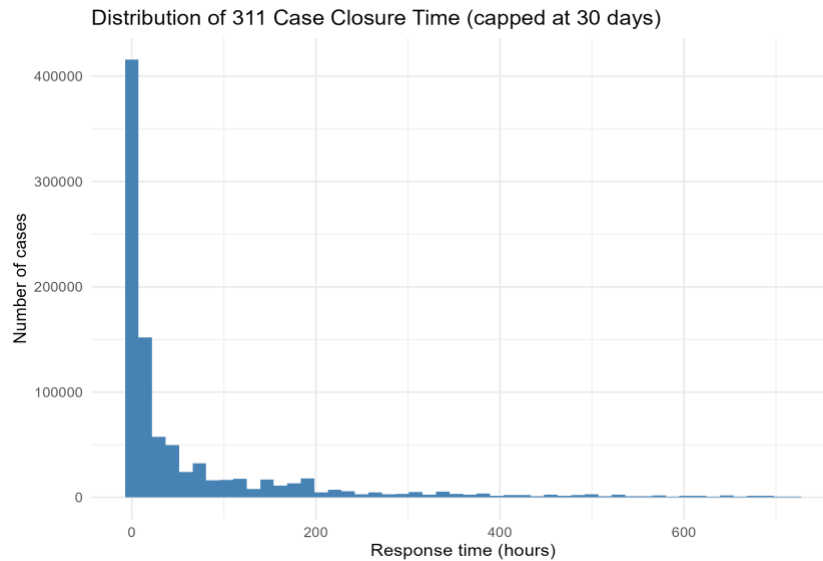
This is mainly a distribution appropriate non-regression check on the overdue rate gap reported. This test directly addresses whether the observed gap in overdue rates (14% vs 15%) is larger than would be expected under independence, without relying on the linear probability assumptions of the logistic model. All regression models were fit in R using the tidyverse, broom, and pROC packages.

Characteristic	Full sample (N = 1,031,658)
Response time (hours), M (SD)	293.9 (954.6)
Case on time, n (%)	884,026 (86%)
Case overdue, n (%)	147,474 (14%)

Characteristic	Higher income (N = 549,957)	Lower income (N = 481,701)	Both (N = 1,031,658)
Response time (hours), M (SD)	290.6 (972.3)	297.6 (934.0)	293.9 (954.6)
Case on time, n (%)	473,368 (86%)	410,658 (85%)	884,026 (86%)
Case overdue, n (%)	76,521 (14%)	70,953 (15%)	147,474 (14%)

These two tables reproduce the full sample and income tier descriptive statistics established in the initial report and now taken from the finalized 1,031,658 case analysis file. Lower income neighborhoods show both a longer mean response time (297.6 vs 290.6 hours) and a higher overdue rate (15% vs 14%) than higher income neighborhoods. Two figures were created with the first one being a distribution of 311 case closure time with a right skew capped at 30 days for visibility, and a figure showing mean response time by neighborhood income tier and year (2015 – 2019). The second figure addresses RQ2 because both income tiers improved substantially in average response time between 2015 and 2019 and the gap between them narrowed sharply after 2016. Lower income neighborhoods began the period noticeably slower than higher income

neighborhoods (363 vs 317 mean hours in 2015) but briefly overtook higher-income neighborhoods by 2018 – 2019. This was when higher income neighborhoods showed a modest uptick (272 to 284 hours) while lower income response times remained essentially flat.



This suggests that the citywide gap narrowed overall but did not fully close and its direction reversed slightly in the most recent year observed.

The two tables below report the full multiple linear regression model predicting response time and the second table reports the full binary logistic regression model predicting overdue status which are both fit on the neighborhood linked analysis file. Department and submission channel are by far the strongest predictors of response time – relative to Inspectional Services, Public Works/Water and Transportation cases close roughly 570 – 600 hours faster while Parks & Recreation and Other City Services cases take 360 – 400 hours longer while holding all other predictors constant.

Term	b	p
(Intercept)	679.87	<0.001
Dept: Mayor's Hotline	-301.47	<0.001
Dept: Other City Services	358.94	<0.001
Dept: Parks & Recreation	459.02	<0.001
Dept: Public Works/Water	-598.37	<0.001
Dept: Transportation	-572.70	<0.001
Channel: Other/Internal	402.43	<0.001
Channel: Phone	60.53	<0.001
Channel: Web/Self-Service	78.43	<0.001
Year 2016	-27.80	<0.001
Year 2017	-43.95	<0.001
Year 2018	-29.94	<0.001
Year 2019	-16.36	<0.001
Season: Spring	-5.22	0.040
Season: Summer	-26.48	<0.001
Season: Winter	25.07	<0.001
Median household income	0.00047	<0.001
Poverty rate	107.93	<0.001
Percent minority	3.26	0.686

This table is a multiple linear regression predicting response time in hours ($N = 1,031,658$) and reference categories: Department = Inspectional Services, Channel = Mobile App, Year = 2015, Season = Fall. The $R^2 = 0.109$, the adjusted $R^2 = 0.109$, $F(18, 1,031,639) = 7,016$, and $p < 0.001$. All years from 2016 – 2019 are significantly faster than 2015, though the year over year improvement was smaller in 2019 than in 2017 (a partial plateau), and winter cases take about 25 additional hours relative to fall which is consistent with snow and storm related workload.

Poverty rate is a significant positive predictor of response time ($b = 107.9$, $p < 0.001$) meaning that moving from a 0% to 100% neighborhood poverty rate is associated with roughly 108 additional hours to close a case, net of department, channel, year, and season. Median household income is also significantly positive ($b = 0.00047$, $p < 0.001$) meaning wealthier neighborhoods show slightly longer response times net of department and channel mix. Percent minority was not a significant predictor once poverty rate and income were included ($p = 0.686$). The equity story is clearer in the logistic model because poverty rate has the largest effect of any predictor ($OR = 4.93$, 95% CI [4.30, 5.66]). A neighborhood at 100% poverty has roughly 4.9 times the odds of an overdue case relative to a neighborhood at 0% poverty. Percent minority is independently significant ($OR = 1.43$, 95% CI [1.36, 1.51]) and median household income shows a small but significant positive association ($OR = 1.0000006$ per dollar). These two models directly answer the first half of RQ1: yes, response times and overdue rates differ systematically by neighborhood poverty and racial composition. The poverty and racial disparities persist and become clearer after controlling for department, channel, year, and season.

Term	Odds Ratio	95% CI
(Intercept)	0.326	[0.297, 0.357]
Dept: Mayor's Hotline	0.005	[0.004, 0.007]
Dept: Other City Services	0.466	[0.448, 0.484]

Term	Odds Ratio	95% CI
Dept: Parks & Recreation	0.191	[0.184, 0.198]
Dept: Public Works/Water	0.272	[0.267, 0.277]
Dept: Transportation	0.268	[0.263, 0.274]
Channel: Other/Internal	0.598	[0.578, 0.619]
Channel: Phone	1.157	[1.141, 1.172]
Channel: Web/Self-Service	0.755	[0.734, 0.776]
Year 2016	0.342	[0.336, 0.349]
Year 2017	0.366	[0.359, 0.372]
Year 2018	0.428	[0.420, 0.435]
Year 2019	0.441	[0.434, 0.449]
Season: Spring	1.359	[1.335, 1.383]
Season: Summer	1.066	[1.047, 1.085]
Season: Winter	1.974	[1.942, 2.007]
Median household income	1.000006	[1.000005, 1.000006]
Poverty rate	4.933	[4.302, 5.657]
Percent minority	1.434	[1.361, 1.510]

This table is a binary logistic regression predicting case overdue status, reported as odds ratios (N = 1,031,500). Null deviance = 846,489 on 1,031,499 df, residual deviance = 778,107 on 1,031,481 df, and AIC = 778,145.

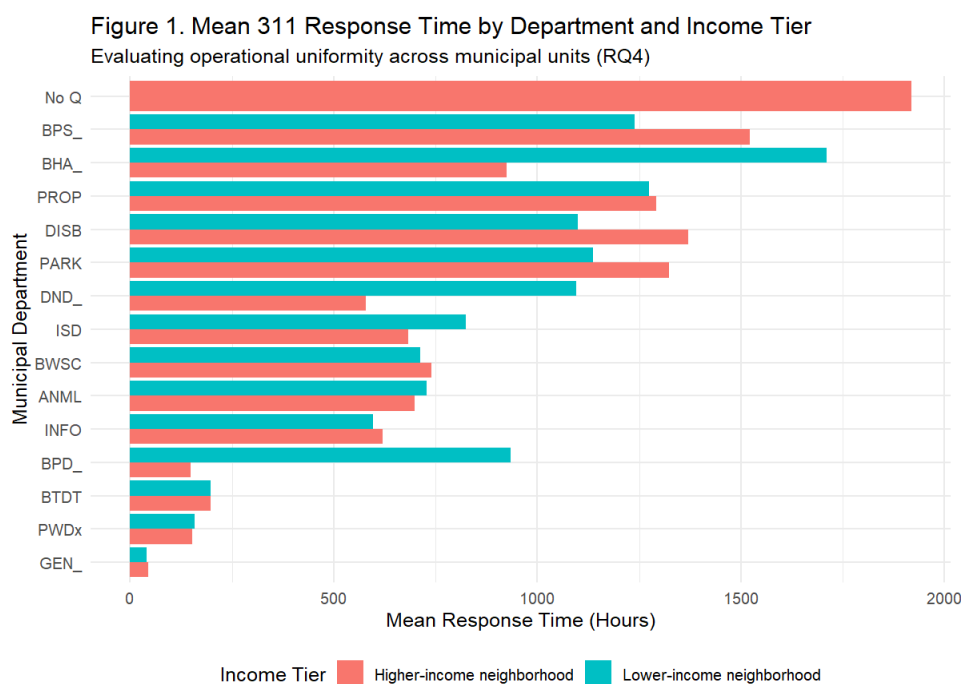
The next table reports selected coefficients from the base and controlled linear models fit for RQ3. In the base model, poverty rate is a marginally significant predictor of response time ($b = -31.59$, $p = 0.063$) while lower income tier remains a significant positive predictor ($b = 10.29$, $p < 0.001$). After adding compliance status and complaint type in the controlled model, the poverty rate coefficient remains negative and becomes more precisely estimated ($b = 031.16$, $p = 0.026$), while the lower income coefficient grows slightly and remains significant ($b = 14.59$, $p <$

0.001). Complaint type itself is highly predictive in terms of relative to the omitted reference category, request such as New Tree Requests ($b = 1,521.64$) and Street Light Longterm Repair ($b = 1,573.27$) takes substantially longer to close. This is compared to requests such as Roadway Repair ($b = -397.98$) which close faster, holding poverty, income tier, and compliance status constant. Overdue cases average about 951 additional hours to close relative to on-time cases ($b = 950.91$, $p < 0.001$), which functions as a validation check on the outcome coding rather than a genuine control for case urgency.

Term	Base model b	Controlled model b	Controlled p
Poverty rate	-31.59 ($p = .063$)	-31.16	0.026
Income tier: Lower-income	10.29 ($p < .001$)	14.59	<0.001
Compliance status: Overdue	—	950.91	<0.001
Complaint type: New Tree Requests	—	1,521.64	<0.001
Complaint type: Street Light Longterm Repair	—	1,573.27	<0.001
Complaint type: Roadway Repair	—	-397.98	<0.001
Complaint type: Sidewalk Repair	—	184.60	<0.001

Selected results from the RQ3 base and controlled linear regression models are shown above. In RQ1 models, which controlled for department, channel, year, and season, poverty rate was a strong positive predictor of delay with $b = 107.9$ in the linear model or $b = 4.93$ in the logistic model. In this case, controlling instead for complaint type and compliance status gives a poverty rate coefficient that is negative. This indicates that the poverty rate effect is sensitive to which set of operational covariates are included. It could suggest that part of RQ1 for the poverty effect may operate through which department handles a case and how it is submitted, rather than through complaint type alone. The income-tier gap is far more stable as lower-income

neighborhoods are associated with significantly longer response times in every specification tested. Compliance status is derived from the same underlying timestamps as the response variable and is not a true measure of case urgency – the $b = 951$ coefficient it receives should be interpreted as a specification limitation rather than a genuine urgency control.



This figure shows Mean 311 response time by municipal department and neighborhood income tier and that the income tier gap in mean response time is not uniform across departments. Within the reference department of Animal Control (ANML), poverty rate is a strong positive predictor of response time

Department	Poverty × Dept (b, p)	Income tier × Dept (b, p)	Net pattern relative to baseline
Boston Housing Authority (BHA)	+7,650.9 (p = .024)	-326.5 (p = .579, ns)	Poverty effect roughly quadruples (net ≈ +10,200); driven by poverty, not the binary income split

Department	Poverty $\times$ Dept (b, p)	Income tier $\times$ Dept (b, p)	Net pattern relative to baseline
Boston Police Dept. (BPD)	-4,852.4 (p = .042)	+1,341.6 (p < .001)	Poverty effect reverses (net $\approx$ -2,303); large, significant income-tier gap emerges instead (net $\approx$ +1,084 hrs slower for lower-income)
Boston Public Schools (BPS)	+6,367.7 (p = .005)	-931.7 (p = .005)	Poverty effect intensifies (net $\approx$ +8,917); income-tier gap reverses, lower-income faster (net $\approx$ -1,190 hrs)
Disability Commission (DISB)	-10,021.9 (p < .001)	-15.6 (ns)	Poverty effect fully reverses (net $\approx$ -7,473); no income-tier heterogeneity
Inspectional Services (ISD)	-3,088.5 (p < .001)	+454.9 (p < .001)	Poverty effect weakens/reverses (net $\approx$ -539); modest income-tier gap remains (net $\approx$ +197 hrs)
Parks (PARK)	-4,797.9 (p < .001)	+335.6 (p = .001)	Poverty effect reverses (net $\approx$ -2,248); modest income-tier gap remains (net $\approx$ +77 hrs)
Neighborhood Development (DND)	+4,332.7 (ns, p = .113)	-15.3 (ns, p = .975)	No significant heterogeneity detected relative to baseline
Property Management (PROP)	-721.3 (ns, p = .303)	+49.9 (ns, p = .636)	No significant heterogeneity detected relative to baseline

This table shows selected department interaction terms from the RQ4 model with the reference department = Animal Control. Net pattern combines the department interaction with the corresponding baseline main effect (poverty rate = 2,549.47, lower income tier = -258.41). This is because the lower income tier on its own is associated with a faster response and the department interaction terms show that this baseline pattern shifts substantially and in some cases, reverses, across departments. The large lower income disadvantage that is visible for BHA is statistically explained by poverty rate rather than by the binary income tier split. The poverty interaction is large and significant while the income tier interaction is not. The opposite pattern

holds for BPD where the poverty interaction reverses sign, but the income tier interaction is large, positive, and highly significant. This means the disparity for police handled requests is better captured by whether a neighborhood is above or below the median income, rather than its poverty rate specifically. BPS shows a significant disparity in both directions simultaneously since poverty widens it and income tier narrows it – this illustrates why a single citywide coefficient, such as what was used in RQ1, can mask offsetting neighborhood effects. By contrast, DND and PROP show no statistically detectable difference at all, meaning their department level bars are not distinguishable from the citywide baseline once sampling variability is accounted for. This helps directly answer RQ4 by saying the equity gap is not uniform across departments and it is statistically concentrated in a subset of operational units (BHA, BPD, BPS, DISB, ISD, and PARK). Even within this subset, the specific socioeconomic driver (poverty rate vs income tier) differs by department.

A chi square test of independence was conducted on the 2 by 2 table of income tier by case status using the counts reporting higher income: 473,368, overdue: 76,521, lower income: 410,658, overdue: 70,953. The association between income tier and overdue status was statistically significant $\chi^2(1, N = 1,031,500 = 139.76, p < 0.001)$. The corresponding effect size was negligible with Cramer's $V = 0.012$ and this indicates that although the overdue rate gap between income tiers (14% vs 15%) is unlikely to be due to chance given the very large sample size, income tier alone explains only a small share of the variation in whether a case is overdue. This is consistent with the RQ1 logistic model saying that poverty rate and percent minority (continuous, more granular predictors) carry most of the explanatory weight while the coarser binary income tier split used for this chi-square test and for descriptive tables captures on a small percent of that same relationship. RQ3, RQ4, and chi-square results reinforce the conclusion that

neighborhood socioeconomic status is a real but modest average predictor of case outcomes citywide. It's true explanatory power only becomes visible once it is either measured continuously (poverty rate RQ1) or allowed to vary by department (RQ4).

RQ1: do response times and overdue rates differ systematically by neighborhood income, poverty, and racial composition, net of department, channel, year, and season? Yes, because in the full linear model, poverty rate remains a significant positive predictor of response time even after controlling for other variables ($b = 107.9$, $p < 0.001$) and in the full logistic model, poverty rate is the single largest predictor of overdue status ($OR = 4.93$). Percent minority is also independently significant in the logistic model ($OR = 1.43$). Median household income shows a small, statistically significant positive association with delay in both models. RQ2: has the gap widened or narrowed over 2015 – 2019? It has narrowed overall but not completely with lower income neighborhoods closing a roughly 46-hour gap with higher income neighborhoods between 2015 and 2017. It essentially matched higher-income response times by 2018 and then pulled slightly ahead of higher income neighborhoods in 2019 as higher income response times ticked back up. The year coefficients corroborate a cityside slowdown in the rate of improvement after 2017 which is a partial plateau. RQ3: does the poverty effect persist and is the counter-intuitive income effect resolved once complaint type and urgency are controlled? The answer is partially and there is an important caveat – in the RQ3 models which control for complaint type and case compliance status, but not department, channel, year, or season, the poverty rate coefficient is negative rather than positive, which is the opposite sign from the RQ1 model. This does not mean poverty rate has no real effect but rather means the direction of the estimated effect depends heavily on which operational covariates are included. The RQ3 specification omits department and channel, which RQ4 shows are these are department specific moderators

of the poverty effect. The income tier gap is stable and positive (slower for lower income neighborhoods) in every specification tested across both reports, including the RQ3 controlled model ($b = 14.59$, $p < 0.001$). The counter intuitive income effect from RQ1 is therefore not fully resolved by complaint type alone and RQ4 suggests it is better explained by which department is handling the case. RQ4: do equity gaps manifest uniformly across departments or are they concentrated in specific operational units? They are concentrated but not uniform because the interaction model identifies statistically significant department specific disparities for BHA, BPD, BPS, DISB, ISD, and PARK, with the specific socioeconomic driver (poverty rate vs binary income tier) differing by department – this was all while DND and PROP show no significant heterogeneity relative to the citywide baseline. This is the most actionable finding of the analysis and shows that equity interventions aimed at Boston's 311 system should be targeted at specific departments rather than treated as a single citywide policy problem.

Conclusion:

The five methods applied conclude that Boston's 311 system is not uniformly inequitable but is definitely not completely fair. A modest, statistically robust neighborhood-based gap in response time and overdue rate exists citywide and it narrowed substantially but not completely between 2015 and 2019 (RQ2). Its apparent driver shifts depending on model specification in a way that complaint type alone does not fully resolve RQ3 and most importantly, it is concentrated in an identifiable subset of departments rather than spread evenly across city operations. The practical recommendation for a municipal audience suggests that oversight and process improvement should be targeted at the departments where there are the most robust disparities which include Boston Housing Authority and Boston Police Department. Departments like Neighborhood Development and Property Management, which show no significant income

or poverty related heterogeneity, do not need the same amount of attention. Future work adding onto this analysis should combine the department interaction approach with the full RQ1 control set (department, channel, year, and season together with complaint type and neighborhood characteristics) into a single specification. This is so that the RQ3 sign reversal can be attributed definitively to complaint type, department, or both. Given the very large number of complaint type categories, a regularized approach such as a LASSO regression would be a natural next step: it would allow the model to retain the most predictive complaint type categories while shrinking the coefficients of rare or unstable categories towards zero, which reduces the risk of overfitting that a fully saturated complaint type model carries otherwise.

Several limitations continue to apply to this analysis, and the first one is the compliance status variable used as a control in the RQ3 controlled model being derived from the same case closure timestamps as the response time outcome. Therefore, its large coefficients reflect definitional overlap rather than an independent urgency measure and a cleaner urgency control (complaint type level priority code if available in the source system) should replace it in any future work. The RQ3 and RQ4 models in this report do not simultaneously include department, channel, year, and season alongside complaint type and the income/poverty variables – so, the sign reversal in the poverty rate coefficient cannot yet be attributed definitively to complaint type mix vs the department and channel level moderation documented in RQ4. The very large number of complaint type categories (over 100) and department interaction terms raises the risk of unstable estimates for rare categories and motivates the LASSO based extension. Several of the wider confidence intervals (DND, PROP) are consistent with comparatively small department by income tier cell sizes rather than a true absence of disparity and should not be over-interpreted as definitive evidence of equity. The department interaction model was estimated on poverty rate

and income tier without the department by channel or department by year structure, so some of the department specific effects may partly reflect omitted seasonal or year-based confounding within departments. Lastly, response time and the BPDA/Census neighborhood characteristics are for a single five-year snapshot that cannot capture within neighborhood change outside of the study period.

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Appendix: R Code

```

5 #Clean canvas ----
6 # clears console
7 cat("\014")
8 # clears global environment
9 rm(list = ls())
10 # clears plots
11 graphics.off()
12 # clears packages
13 try(p_unload(p_loaded()), character.only = TRUE), silent = TRUE)
14 # disables scientific notation for entire R session
15 options(scipen = 100)
16
17 # 1. Load Required Libraries
18 library(tidyverse)
19 library(broom)
20 library(pROC)
21
22 # 2. Load the Cleaned Dataset
23 # Ensure your working directory contains the merged/cleaned file
24 df <- read.csv("d311_clean_analysis_file.csv", stringsAsFactors = FALSE)
25
26 # 3. Feature Engineering & Transformations (Using actual column names)
27 df <- df %>%
28   mutate(
29     # Convert grouping and control variables to factors
30     income_group = as.factor(income_group),
31     department = as.factor(department),
32     complaint_type = as.factor(type),
33
34     complaint_reason = as.factor(reason),
35     # Use 'on_time' or 'overdue' as the compliance/urgency control for RQ3
36     compliance_status = as.factor(on_time)
37   )
38
39 # 4. Summary Statistics Table by Income Tier (Mean and SD format matching course examples)
40 summary_table <- df %>%
41   group_by(income_group) %>%
42   summarise(
43     N = n(),
44     Mean_Response_Hours = round(mean(response_hours, na.rm = TRUE), 2),
45     SD_Response_Hours = round(sd(response_hours, na.rm = TRUE), 2),
46     Mean_Poverty = round(mean(poverty_rate, na.rm = TRUE), 2),
47     SD_Poverty = round(sd(poverty_rate, na.rm = TRUE), 2)
48   )
49
50 cat("\n=== Table 1. Summary Statistics by Income Tier ===\n")
51 print(summary_table)
52 write.csv(summary_table, "Table1_SummaryStats_IncomeTier.csv", row.names = FALSE)
53
54 # 5. RQ3 Analysis: Controlling for Complaint Type and Compliance/Urgency
55 # Model 1: Base model (Poverty & Income only)
56 m_base <- lm(response_hours ~ poverty_rate + income_group, data = df)
57
58 # Model 2: Controlled model (Adding compliance status and complaint type)
59 m_controlled <- lm(response_hours ~ poverty_rate + income_group + compliance_status + complaint_type, data = df)
60
61 cat("\n=== RQ3: Base Model Summary ===\n")

```

```

61 print(summary(m_base))
62
63 cat("\n=== RQ3: Controlled Model Summary ===\n")
64 print(summary(m_controlled))
65
66 # Export RQ3 regression results as a structured CSV table for reporting
67 rq3_results <- bind_rows(
68   tidy(m_base) %>% mutate(Model = "Base Model"),
69   tidy(m_controlled) %>% mutate(Model = "Controlled Model")
70 ) %>%
71   mutate(across(where(is.numeric), ~ round(.x, 4)))
72
73 write.csv(rq3_results, "Table2_RQ3_Regression_Results.csv", row.names = FALSE)
74
75 # 6. RQ4 Analysis: Departmental Differences (Interaction Models)
76 # Model 3: Interaction between Income/Poverty tiers and Municipal Departments
77 m_dept_int <- lm(response_hours ~ (poverty_rate + income_group) * department, data = df)
78
79 cat("\n=== RQ4: Department Interaction Model Summary ===\n")
80 print(summary(m_dept_int))
81
82 # Export RQ4 interaction results
83 rq4_results <- tidy(m_dept_int) %>%
84   mutate(across(where(is.numeric), ~ round(.x, 4)))
85
86 write.csv(rq4_results, "Table3_RQ4_Interaction_Results.csv", row.names = FALSE)
87
88 # 7. Visualization & Export (High-Resolution PNG)
89
90 plot_data <- df %>%
91   filter(!is.na(department) & !is.na(income_group) & !is.na(response_hours)) %>%
92   group_by(department, income_group) %>%
93   summarise(mean_response = mean(response_hours, na.rm = TRUE), .groups = 'drop')
94
95 p1 <- ggplot(plot_data, aes(x = reorder(department, mean_response), y = mean_response, fill = income_group)) +
96   geom_bar(stat = "identity", position = "dodge") +
97   coord_flip() +
98   labs(
99     title = "Figure 1. Mean 311 Response Time by Department and Income Tier",
100     subtitle = "Evaluating operational uniformity across municipal units (RQ4)",
101     x = "Municipal Department",
102     y = "Mean Response Time (Hours)",
103     fill = "Income Tier"
104   ) +
105   theme_minimal(base_size = 12) +
106   theme(legend.position = "bottom")
107
108 # Save plot to file
109 png("Figure1_ResponseTime_by_Department.png", width = 1200, height = 900, res = 150)
110 print(p1)
111 dev.off()
112
113 cat("\n=== Analysis complete. All tables and figures exported successfully. ===\n")

```